장서현
20253254096

IIE 8560 Fall 2025

Homework #1

Due: September 26th 18:00

1. Consider samples $x_1, \dots, x_n$ from a Gaussian random variable with known variance σ^2 and unknown mean μ. We further assume a prior distribution (also Gaussian) over the mean, $\mu \sim N(m, s^2)$, with fixed mean m and fixed variance s^2. Thus, the only unknown is μ. Calculate the MAP estimate $\hat{\mu}_{MAP}$. You can state the result without proof.

$$\hat{\mu}_{MAP} = \frac{ns^2\bar{x} + \sigma^2 m}{ns^2 + \sigma^2}$$

$$\underset{\mu}{\operatorname{argmax}} \left[\ln(p(data|\mu)) + \ln(p(\mu)) \right]$$

$$\ln(p(data|\mu)) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2 \quad ①$$

$$\ln(p(\mu)) = -\frac{1}{2s^2}(\mu-m)^2 \quad ②$$

$$L(\mu) = ① + ②$$

$$\frac{\partial L}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) - \frac{1}{s^2}(\mu-m) = 0$$

$$\frac{\sum x_i}{\sigma^2} + \frac{m}{s^2} = \mu\left(\frac{n}{\sigma^2} + \frac{1}{s^2}\right)$$

$$\frac{n\bar{x}}{\sigma^2} + \frac{m}{s^2} = \mu\left(\frac{n}{\sigma^2} + \frac{1}{s^2}\right)$$

$$\hat{\mu}_{MAP} = \frac{ns^2\bar{x} + \sigma^2 m}{ns^2 + \sigma^2}$$

2. Suppose we sample $x_1, \dots, x_n \sim N(\cancel{x}\,\mu, \sigma^2)$, where μ is a known constant. Derive an expression for the MLE for σ^2 in this case. Is it unbiased?

$$L(\sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right)$$

$$\ln L(\sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

$$\frac{\partial}{\partial\sigma^2}\ln L(\sigma^2) = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i-\mu)^2 = 0$$

$$n\sigma^2 = \sum_{i=1}^{n}(x_i-\mu)^2$$

$$\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2$$

unbiased

$$E[\hat{\sigma}^2_{MLE}] = E\left[\frac{1}{n}\sum_{i=1}^{n}(x_i-\mu)^2\right]$$

$$= \frac{1}{n}\sum_{i=1}^{n}E[(x_i-\mu)^2]$$

$$\mu = E[x_i]$$

$$= \frac{1}{n}\sum_{i=1}^{n}E[(x_i-E(x_i))^2]$$

$$= \frac{1}{n}\sum_{i=1}^{n}Var(x_i) = \frac{1}{n}\cdot n\cdot\sigma^2 = \sigma^2$$

3. After your yearly checkup, the doctor has bad news and good news. The bad news is that you tested positive for a serious disease, and that the test is 99% accurate (i.e., the probability of testing positive given that you have the disease is 0.99, as is the probability of testing negative given that you don't have the disease). The good news is that this is a rare disease, striking only one in 10,000 people. What are the chances that you actually have the disease? (Show your calculations as well as giving the final result.)

D: event of one having the disease $\rightarrow P(D) = 1/10000 = 0.0001$

D': event of one not having the disease $\rightarrow P(D') = 1 - P(D) = 0.9999$

T+ : event of one getting tested positive $\rightarrow P(T^+|D) = 0.99$

$\rightarrow P(T^+|D') = 1 - P(T^+|D) = 0.01$

$$P(T^+) = P(T^+|D) \times P(D) + P(T^+|D') \times P(D')$$
$$= 0.99 \times 0.0001 + 0.01 \times 0.9999 = 0.010098$$

$$P(D|T^+) = \frac{P(T+|D) \times P(D)}{P(T+)} = \frac{0.99 \times 0.0001}{0.010098} = 0.0098$$

$\therefore 0.98\%$

4. Conjugate Prior for Bernoulli Likelihood. Consider the following model:

Likelihood: $x_i \mid \theta \sim Bernoulli(\theta)$, for $i = 1, \ldots, n$

Prior: $\theta \sim Beta(\alpha, \beta)$

(a) Show that the prior is conjugate to the likelihood.

(b) Derive the posterior distribution $p(\theta \mid x)$, and specify its form and parameters.

(a)

Likelihood

$: p(x|\theta) = \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i}$

$= \theta^{\Sigma x_i}(1-\theta)^{n-\Sigma x_i}$

Prior

$: p(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)+\Gamma(\beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1}$

$p(\theta|x) \propto p(x|\theta) \cdot p(\theta)$

$\propto [\theta^{\Sigma x}(1-\theta)^{n-\Sigma x}] \cdot [\theta^{\alpha-1}(1-\theta)^{\beta-1}]$

$\propto \theta^{\Sigma x+\alpha-1}(1-\theta)^{n-\Sigma x+\beta-1}$

Since the posterior is in the same family as the prior, the Beta prior is conjugate to the Bernoulli likelihood.

(b)

$p(\theta|x) \propto \theta^{\Sigma x+\alpha-1}(1-\theta)^{n-\Sigma x+\beta-1}$

$\propto \theta^{\alpha'}(1-\theta)^{\beta'}$

$\downarrow$

Beta Distribution

posterior

$: \theta|x \sim Beta(\sum_{i=1}^{n} x_i + \alpha, n - \sum_{i=1}^{n} x_i + \beta)$